Affine Geometry and Fast Cauchy Transforms for Subexponential SAT

A smaller certified contraction exponent, exact local transforms, and a comparison with the newly published Anthropic formalization

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October 26, 2026 (Pacific date)

Abstract

We review ProveIt's four rounds of subexponential-SAT refinements at commit `a7ae5ab55f`, including all nine newly placed round-4 packets, and the Anthropic formal-math revision supplied during this review. A new finite geometric construction lowers the strongest explicit generic cached-contraction bound from time coefficient $30.891018920366\ldots$ to $30.413348726348439\ldots$, and from space coefficient $30.753912832055\ldots$ to $29.466030256594790\ldots$, on the normalized scale $s/\sqrt{\lceil \log_2 s \rceil}$. Four parallel classes of the affine plane $\mathbb{F}_3^2$ and a thirteen-plan cyclic construction provide exact correlation certificates. A fully paid recursion includes every child entry. Separately, we give precision-safe fast applications of the normalized Cauchy feature matrix. At odd interpolation nodes the application reduces to one convolution; exact tests and matched small benchmarks accompany it. These are new improvements relative to the inspected corpus. They do not establish a smaller logarithmic order for unrestricted SAT. We explain why a direct combination with Anthropic's clique algorithm does not by itself remove that obstacle, and isolate a conditional graded-algebra route and an obstruction to its most immediate construction.

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1 Results and their scope

The strongest new result is a strictly smaller certified exponential upper bound for the same generic normalized contraction problem used by the most recent packets. Set $\lambda = \lceil \log_2 s \rceil$ and $E(s) = s/\sqrt{\lambda}$. The exact coefficients are

$$C_T = \frac{\pi}{\ln 2}\left(\frac{4}{9}\sqrt{\frac{2000000}{15773}} + \frac{160\sqrt{14}}{351}\right) = 30.413348726348439\ldots, \tag{1}$$

$$C_S = \frac{\pi}{\ln 2}\sqrt{\frac{2000000}{47319}} = 29.466030256594790\ldots. \tag{2}$$

Under the interface of Section 3, time is $(s+\mathcal{L})^{O(1)}2^{(C_T+o(1))E(s)}$ and space is $(s+\mathcal{L})^{O(1)}2^{(C_S+o(1))E(s)}$. Eventual coefficients 30.414 and 29.467 are certified by exact rational interval calculations. The square-root ceiling changes no leading constant relative to the $s/\sqrt{\log_2 s}$ scale used by some other packets.

Generic normalized certificate	Time	Space
Best reviewed polynomial-space branch [9]	46.184846	polynomial
Best reviewed cached branch [11]	30.891019	30.753913
New menus, earlier three-stage schedule	30.512927	29.487537
New menus, final four-stage schedule	**30.413349**	**29.466030**

The new cached time coefficient is about 1.5463% smaller and its space coefficient about 4.1877% smaller than the strongest displayed comparable generic cached result. These percentages compare coefficients, not observed run times. If U_{old} and U_{new} denote the explicit upper-bound expressions, then

$$\log_2 \frac{U_{\mathrm{old}}}{U_{\mathrm{new}}} = (0.477670194017653\ldots + o(1))\,E(s).$$

This is a meaningful asymptotic improvement of the upper-bound function; it is not a lower bound on the time consumed by the older algorithm.

The new idea. The earlier best cached proof selected among three contraction trees and enumerated all tuples of their operation supports. We replace that small unstructured menu with a finite geometric menu and a pairwise correlation certificate. The nine-group construction uses the four parallel classes of $\mathbb{F}_3^2$; the thirteen-group construction uses cyclic shifts of one explicit seed. The certificate controls every feasible allocation of mode weights, including table construction. It pays for every actual recursive child-factor evaluation. This is a corpus-relative contribution; averaging, finite geometry, and fast matrix multiplication are classical tools.

A second, local asymptotic improvement. The normalized Cauchy matrix need not be materialized to compute its source linear forms. We give guarded product-tree algorithms for matrix-vector application and its transpose. At odd interpolation nodes, the features have a diagonal–Toeplitz–diagonal representation after known powers of two are cleared. All source forms then require one convolution. In the scalar ring this admits bit cost $\widetilde{O}(N(p+N))$, hence $\widetilde{O}(Np)$ for $p \geq N$, including the precision guards. The reviewed streamed application uses quadratic feature work. The prototype's one-convolution specialization is faster on all five reported fixtures; the general product-tree prototype is slower at those sizes, and those regressions are retained.

The requested larger breakthrough remains open. The global SAT order remains

$$\mathrm{poly}_k(n,m)2^{O_k(n\log\log n/\sqrt{\log n})}$$

in the unrestricted fixed-width case and $2^{O_{k,D}(n/\sqrt{\log n})}\,\mathrm{poly}(n,m)$ at fixed clause density. No $n/\log n$ algorithm is proved here. The complete deliverable separates the new proved finite lemmas, imported analytic and arithmetic ingredients, tested implementation, and genuinely conditional ideas. No end-to-end SAT solver or new Lean formalization is claimed.

The coefficients above are per normalized Boolean wire s. They cannot be inserted unchanged in an exponent written in SAT variables n. The existing density-sensitive and low-degree branches remain in the available portfolio; they have different premises. Nor can a numerical coefficient be claimed to dominate an unspecified constant in the new Anthropic big-O bound. These distinctions are necessary to make the comparison mathematically meaningful.

2 Recent commits, the full comparison set, and the supplied PR

2.1 Pinned review and incoming arrivals

The review is frozen at ProveIt commit `a7ae5ab55f9627f5af7bf77443b3c34ef8ef1be4`. It was committed at 20:15:51 Pacific on October 26, 2026. The relevant progression is:

Commit	Relevant action
`2b925d8800`	Places the seven first-round packets and primary manuscripts.
`3e09608991`	Vendors the OpenAI manuscript and its four LaTeX sources.
`6d3adc4506`	Places eight round-2 packets.
`450252e9d8`	Places five round-3 packets and restores two replay scripts.
`b9941a7396`	Commits the nine new incoming round-4 archives.
`a7ae5ab55f`	Places those nine packets under SubexponentialSAT.
`7de9d09fec`	Links the Anthropic formalization; no mathematical packet changes.

At the review pin, `docs/incoming/` contains its README only. Its nine recent SAT arrivals have already been relocated into the research tree. They are included here; reviewing only the earlier fifteen or twenty packets would give an obsolete benchmark. The review covered the 29 packet summaries and status declarations, with detailed examination of the source interfaces and leading proofs used in the new comparison. It did not independently reprove every statement in all 29 packets.

The final branch check observed `7de9d09fecd17de776d712835ecb2011205c531d`, committed at 20:20:40 Pacific. We inspected its complete relevant diff: it updates the README to link the Anthropic development and adds no new proposal or complexity bound. The mathematical corpus remains pinned at `a7ae5ab55f9627f5af7bf77443b3c34ef8ef1be4`; the documentation-only follow-up is recorded separately.

The accompanying source manifest records immutable source paths, Git blob IDs and SHA-256 hashes. The checkout is unmodified. Historical archives and upstream manuscripts are linked rather than duplicated in this deliverable.

2.2 Inventory of the 29 refinement packets

In the following table, PS denotes polynomial space. Cached coefficients permit subexponential space. Unless qualified, numbers are generic normalized-wire time coefficients reported by the corresponding packet, with that packet's stated imported dependencies. This inventory summarizes

the indexed corpus [1]; the new result uses the actual leading interfaces, not just the printed decimal numbers.

Packet	Relevant result or limitation
`eth_contraction_improvements`	Sharper Hermite taper and streamed extraction; PS 648.441.
`cauchy_mode_improvement`	Factorial modes; ratio $1/\sqrt{3}$ in the mode-enumeration constant.
`3sat_precision_refinement`	Precision $24MD/u \to 6MD/u$; the stronger $n/\log n$-type rates need an unproved summation algorithm.
`sat_density_improvement`	Density-sensitive order envelope with $\min\{\log(2 + m/n), \log\log n\}$.
`eth_density_refinement`	Independent intermediate-density improvement and constant refinements.
`3sat_refinement`	Scalar parity lift; second-contraction factorization remains missing.
`3sat_asymptotic_analysis`	Scoped taper, rank and degree barriers; conditional interfaces remain unconstructed.
`eth_asymmetric_packet_balancing`	PS 76.909; asymmetric near-unit padding.
`3sat_round2_refinement`	PS 84.066; independent source and actual-domain audit.
`eth_balanced_packing`	PS 94.748; thin precision buffers and near-unit padding.
`eth_high_moment`	PS 94.748; degree-sensitive extensions.
`eth_matching_refinement`	Third independent 94.748 bound through matching-based packing.
`eth_packet_refinement`	PS 95.664; improved pair loads.
`eth_recursive_refinement`	PS 133.207; refined Cauchy residue classes.
`eth_joint_cauchy`	Full-block slope-four modes; PS 319.686, cached 225.597. Its mode identity is used by later improvements.
`sat_adaptive_contraction`	PS 55.709; cached 36.750; modular Cauchy kernels.
`sat_frontier_improvements`	PS 63.857, cached 38.484; variable arity and quadratic feature construction.
`subexponential_sat_frontier`	PS 63.852, cached 37.930; exact labeling and precision peeling.
`subexponential_sat_kernel_improvements`	PS composition 66.605; Smith, streamed and projective coefficient kernels.
`subsat_contraction_tree`	Cached 35.355; recursive table merging and a rational-matrix-to-ring bridge.
`sat_adaptive_bounds`	Best generic PS 46.184846; complete binary flags. Low-degree cached 13.379649 under stronger assumptions.
`sat_adaptive_tree_certificate`	Best generic cached 30.891019 time and 30.753913 space; adaptive three-plan support certificates.
`sat_adaptive_frontier`	PS 50.30469; cached time/space 31.25240.
`sat_adaptive_matrix_partition`	PS 50.21043; cached 34.32528.

Packet	Relevant result or limitation
`subexponential_sat_adaptive_four`	PS 50.39373; cached 32.91581, or 32.85635 with the stated stronger matrix backend.
`subsat_rectangular`	PS 53.97753; cached 33.99389; precision and table barriers.
`eth_adaptive_exact_kernels`	PS 55.33162, cached 35.07061; sparse-support scoring and fused feature recurrences.
`3sat_round3_frontier`	PS 63.75736; low-degree PS 23.66617; centered extraction kernels.
`eth_adaptive_schedules`	PS 63.78117, cached 37.59170; an incremental matching-potential kernel.

All these packets retain the unrestricted SAT order. The strongest established density-sensitive expression in the corpus is

$$\mathrm{poly}_k(n,m)\,2^{O_k\left(\frac{n}{\sqrt{\log n}}\min\{\log(2+m/n),\log\log n\}\right)}. \tag{3}$$

For example, $m=\Theta(n\log\log n)$ yields $n\log\log\log n/\sqrt{\log n}$ in the exponent. This is an intermediate-density improvement, not a smaller unrestricted order. Conditional targets are not established algorithms that a new numerical coefficient must beat.

2.3 What the Anthropic revision changes

During the review, the user supplied PR 31 of `anthropics/formal-math` at exact revision

`1a6eee293b9e00cb432eef6018a21f523623a9af`.

That revision publishes `eth/`, including the paper, the standalone theorem statements, the proof entry point, the proof tree, and checking instructions [4]. Accordingly, the earlier description of this formalization as unpublished is stale; ProveIt's final documentation-only commit also corrects it. This is a change in source availability and inspectability, not a new smaller exponent stated in the PR.

`MainResults.lean` states the existence of one finite-alphabet Mathlib multi-stack Turing machine and one constant C, both quantified before k,n and the formula. Under $k\ge 1$, $n\ge 4$, $m\ge 2$, no repeated clause-list, and occurrence of every declared variable, its bound is

$$2^{C\min\{km/\sqrt{\log_2 m},\,k^2 n\log_2\log_2 n/\sqrt{\log_2 n}\}}.$$

`Solution.lean` supplies unconditional theorem terms through `CountedRoute.mainTheoremHolds`; the ETH statement follows through the named renaming and decision wrappers. There are 1,052 Lean files in the inspected revision. The two actual `sorry` terms found by the source scan are the intentionally unproved comparison statements in `Challenge.lean`; the proof entry point does not import that file. The comparator configuration lists the standard axioms `propext`, `Classical.choice`, and `Quot.sound`, and requests `nanoda` checking.

These are source-inspection findings. We did not install the pinned Lean toolchain, rebuild the 1,052-file development, or replay its kernel checks. A text scan for `sorry` is not a substitute for such a build. The PDF at the supplied revision is not byte-identical to ProveIt's earlier

`claude-2o-n-3sat/eth.pdf`; both hashes are recorded. We therefore read the supplied paper directly instead of assuming the earlier PDF was the same document.

The newly available source is relevant to the mathematical search. Its clique algorithm admits arbitrary integer-valued high-degree node tables, and its sharp decomposition uses a graded operator algebra with partition-number growth. Section 7 examines both the possible composition with the Cauchy method and the growth-rate suggestion in its Appendix H.

3 The common contraction interface

3.1 What is inherited and what is changed

The root problem is the exact contraction of a bipartite Boolean tensor network with at most s wires, at most polynomially many vertices, and maximum degree at most $\lceil s^{9/10} \rceil$. Its description length is $\mathcal{L}$. Initial entries lie in

$$(\mathbb{Z}/2^{P_0}\mathbb{Z})[V]/(V^\ell), \qquad 1 \le P_0, \ell \le s,$$

and uniform entry evaluation has time $(s+\mathcal{L})^{O(1)}2^{O(s^{9/10})}$ and polynomial space, with fixed constants. The desired contraction is

$$Z = \sum_{x\in\{0,1\}^E} \prod_v T_v(x|_{\delta(v)}).$$

We use the exact deformation, integral evaluations, tapered Hermite recovery, and immutable creation-precision convention of the source and refinements [3, 7, 8, 9, 11]. A retained tensor is evaluated at its creation precision and its canonical coordinates are reused; the canonical-lift map is never assumed to preserve multiplication or addition. Our new global result changes the summation of an already exact finite expression. Its algebraic validity does not depend on experimental timings.

The auxiliary ring is a finite free extension

$$\mathcal{A}_p = (\mathbb{Z}/2^p\mathbb{Z})[U,V]/(f(U), V^\ell),$$

where f is monic of degree $O(\log s)$ with irreducible reduction modulo two. All precisions and coordinate lengths remain polynomial in s. Arithmetic in this ring therefore introduces fixed polynomial factors.

3.2 Packing, modes, and exact list cardinalities

At depth d let u_d be the taper range, let $B = \lceil 6\log_2 \lambda \rceil$ for $\lambda = \lceil \log_2 s \rceil$, and put

$$a_d = \sum_{j\le d}(u_j + B), \qquad L_d = 2^{a_d}, \qquad Q_d = r_d^{a_d}.$$

For a retained interpolation node of valuation $i < u_d$, set $D = L_d/2^i$. The imported sharp precision estimate is

$$p_t \le (6 + o(1))MD/u_d. \tag{4}$$

Nearly lossless asymmetric packing makes each group child have at most $(2/r_d + o(1))M$ wires. Sufficient packing conditions, suppressing fixed polylogarithmic slack, are

$$Q_d R/M = o(1), \quad Q_d^2 C(\log s)^2/M = o(1), \quad L_d Q_d^2/M = o(1). \tag{5}$$

At the root both degree bounds are $\lceil s^{9/10} \rceil$; subsequently one may take retained-side degree $R \le \lceil s^{9/10} \rceil$ and fresh-side degree $C \le 2L_{d-1}$. The concrete schedule below verifies all these conditions.

Full-orientation Cauchy modes have weights at least $1 + 4q - 2\,\mathrm{popcount}(q)$ after lowering the common positive valuation offset. There are at most $(2/r^2 + o(1))M/D$ modes of each index on a prescribed directed pair. Enumerate exact pair-weight arrays $b = (b_{gh})$ whose sum is less than p_t. The alphabet r is fixed, so the number of these arrays is polynomial in s. Let $\mathcal{I}_{gh}$ be the finite list of mode tuples of the specified exact weight. Arrays with an empty list contribute zero. For every remaining array the expression is exactly

$$\sum_{(\xi_{gh}) \in \prod_{g \neq h} \mathcal{I}_{gh}} \prod_{g=1}^{r} Z_g((\xi_e)_{e \ni g}). \tag{6}$$

Each Z_g entry is a genuine child-network contraction, not a free oracle.

List sizes and rank/unrank operations follow from

$$N(j, b) = N(j-1, b) + N(j, b - w_j), \quad N(0, 0) = 1,$$

with zero for negative budgets and for $N(0, b)$ with $b > 0$. Budgets, table dimensions, and bit lengths of counts are polynomial in s. This makes the selectors used later explicit exact algorithms.

3.3 The entropy estimate

Write $h_{gh} = \log_2 |\mathcal{I}_{gh}|$ and

$$\kappa = \frac{\pi\sqrt{2}}{\ln 2}, \qquad \mathcal{K}(M, u) = \kappa M/\sqrt{u}.$$

The common weighted estimate is the following.

Lemma 3.1 (Imported weighted pair entropy)**.** *For every fixed nonnegative directed weight vector z,*

$$\sum_{g \neq h} z_{gh} h_{gh} \le \left(\frac{\|z\|_2}{r} + o(1) \right) \mathcal{K}(M, u).$$

The error is uniform over the active nodes and any fixed finite collection of weight vectors used below.

Proof. The argument is included to expose the constant being used. The generating product for one mode family satisfies

$$\sum_{q \ge 0} -\ln(1 - e^{-\theta(1+4q-2\,\mathrm{popcount}(q))}) = \frac{\pi^2}{24\theta} + O(\log^2(2/\theta)).$$

The weight-counting function is $x/4 + O(\log(x+2))$; partial summation and $\int_0^\infty -\ln(1-e^{-x})\,dx = \pi^2/6$ give this estimate. For positive z_e use the parameter θ/z_e and multiply the logarithm of its bound by z_e. Summing yields leading terms

$$\theta p_t + \frac{\pi^2}{24\theta} \frac{(2 + o(1))M}{r^2 D} \sum_e z_e^2.$$

Optimizing and using (4) gives the claim after division by $\ln 2$. The saddle has scale $\sqrt{u}/D$; the additive guard makes the error $o(M/\sqrt{u})$ uniformly. Zero coordinates are omitted. A fixed finite collection permits taking the largest of its error constants. □

3.4 Cached matrix operations over the actual ring

For disjoint blocks A, B and their complement O, let H_1, H_2, H_3 be the logarithms of the products of index sizes on A–O, A–B, and B–O. The block-table merge is rectangular matrix multiplication. Square blocking with any rational-field matrix exponent strictly below $12/5$ gives exponent

$$\mu(H) = H_1 + H_2 + H_3 - \frac{3}{5}\min(H_1, H_2, H_3) = \max_j \left(\frac{2}{5} H_j + \sum_{k \neq j} H_k \right). \tag{7}$$

The cited matrix result is stronger than the inequality needed here [14]. We retain the exact coordinate/CRT bridge already proved in the cached-tree packet [10]. A brief explanation is useful: expand the coefficient ring into polynomially many coordinates, lift coordinates to integers, and perform the integer coordinate matrix products modulo enough nonexceptional odd primes. A fixed rational bilinear matrix algorithm has only finitely many denominator primes to exclude. CRT reconstructs integer products before applying the ring's structure constants and reducing modulo 2^p. Reconstruction lengths are polynomial in s and the logarithms of matrix dimensions. No nonunit of $\mathcal{A}_p$ is inverted in this argument.

The matrix bound dominates every input and output table size because $\mu(H) \geq H_j + H_k$ for distinct j, k. Sequential rectangular blocking and the fixed bilinear algorithm use a polynomial factor times that storage. Reshaping, indexing, and sorting also cost only a polynomial factor in record and index lengths. The verification code does not implement the theoretical fast matrix backend; that backend is an explicit imported dependency.

4 Affine and cyclic menus with pairwise correlation certificates

4.1 Operation supports and exact selection

For an unordered pair $e = \{g, h\}$ put $g_e = h_{gh} + h_{hg}$. A plan partitions the ordered group labels into three nonempty blocks. Within each block, recursively split its ordered list after $\lfloor k/2 \rfloor$ elements and merge the two boundary tables. Finally contract the three block tables. This gives $r - 3$ binary merges and one final triangle, hence $r - 2$ operations.

Multiplying each of the three linear forms in (7) by five gives a support mask $v \in \{0, 2, 5\}^{\binom{r}{2}}$. Its value is zero inside a class of the operation's tripartition, two on the discounted class pair, and five on the other two class pairs. Let $\mathcal{V}_j$ contain all $3(r-2)$ masks of plan j. After singleton caching, plan j has local operation exponent

$$F_j(g) = \frac{1}{5} \max_{v \in \mathcal{V}_j} v \cdot g.$$

All matrix operations, not only the last triangle, are included.

Theorem 4.1 (Pairwise correlation certificate)**.** *For q fixed plans on r groups, let D_{ij} be any explicit upper bound on $v \cdot w$ for $v \in \mathcal{V}_i, w \in \mathcal{V}_j$, and put $S = \sum_{i,j} D_{ij}$. Selecting the least-cost plan gives local time and storage at most*

$$\mathrm{poly}(s) 2^{(b_r + o(1))\mathcal{K}(M,u)}, \qquad b_r^2 = \frac{2S}{25 q^2 r^2}.$$

The selector is exact and polynomial time in the original size parameter.

Proof. For the given nonnegative g, choose a maximizing v_i from each menu. Then

$$\min_i F_i(g) \le \frac{1}{5q}\left(\sum_i v_i\right)\cdot g, \qquad \left\|\sum_i v_i\right\|_2^2 = \sum_{i,j} v_i \cdot v_j \le S.$$

Give both directed orientations of edge e the weight $(\sum_i v_i)_e/(5q)$. Its squared norm is at most $2S/(25q^2)$. Lemma 3.1 proves the time estimate. The possible choices of supports form a fixed finite set, so the error remains uniform although the selected plan depends on the input. No minimax equality is assumed.

For selection, compute the exact positive integers

$$P_i = \max_{v\in\mathcal{V}_i} \prod_{\{g,h\}} \left(|\mathcal{I}_{gh}|\,|\mathcal{I}_{hg}|\right)^{v_{\{g,h\}}}$$

and choose the smallest. Comparing them is equivalent to comparing $F_i(g)$; no approximate logarithm is needed. Their bit lengths are polynomial and q, r are fixed. Finally every table is an input or output of a selected operation, whose exponent dominates its size by (7). A fixed number of tables introduces only a constant factor. The time to evaluate the singleton entries is charged separately. □

The earlier support-tuple method has an exponential dependence on the number of plans, even though this remains a constant asymptotically. The new certificate requires q^2 pair comparisons. It bounds the simultaneous sum of supports rather than asserting that all pairwise maxima can occur at once. Such an assertion would be unnecessary and generally false.

4.2 Four parallel classes on nine points

Label the groups by $x \mapsto (x \bmod 3, \lfloor x/3 \rfloor) \in \mathbb{F}_3^2$. For each of the four functionals

$$(a,b) \in \{(1,0),(0,1),(1,1),(1,2)\},$$

use its three level sets as the final blocks. Within each level set use increasing integer order. These four plans are

$$\begin{aligned}
&(0,3,6) \mid (1,4,7) \mid (2,5,8),\\
&(0,1,2) \mid (3,4,5) \mid (6,7,8),\\
&(0,5,7) \mid (1,3,8) \mid (2,4,6),\\
&(0,4,8) \mid (1,5,6) \mid (2,3,7).
\end{aligned}$$

Each plan has 21 masks. Exact pairwise maxima are

$$D_{ii} = 486, \qquad D_{ij} = 288 \quad (i \ne j).$$

Thus $S = 4\cdot 486 + 12\cdot 288 = 5400$ and

$$b_9^2 = \frac{1}{3}. \tag{8}$$

This can be checked by 7,056 mask-pair dot products, not by sampling entropy vectors. The independent implementation represents each support as $v = 5\mathbf{1}_C - 3\mathbf{1}_D$ and uses

$$v \cdot w = 25|C\cap C'| - 15(|D\cap C'| + |C\cap D'|) + 9|D\cap D'|. \tag{9}$$

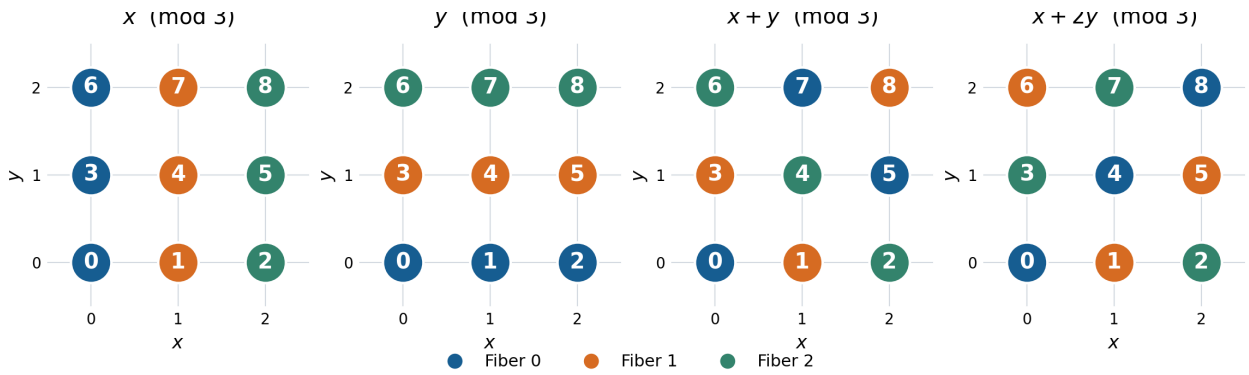


Figure 1: The four parallel classes of $\mathbb{F}_3^2$. In each panel, colors denote the three fibers of the indicated linear functional, with arithmetic modulo three. Labels are the group numbers. Each fiber is one final block; straight Euclidean lines are not required to display the modular wraparound.

There is also a geometric explanation for the largest off-diagonal value. Two distinct directions have a one-point intersection between every pair of their lines. For the final triangles, the two crossing edge sets have intersection size 18; either discounted set meets the other crossing set in 6 edges, and the discounted sets meet in 2. Equation (9) then gives $25 \cdot 18 - 15(6+6) + 9 \cdot 2 = 288$. For completeness, the exact off-diagonal bounds by operation type are

	$(1,1,7)$	$(1,2,6)$	$(3,3,3)$
$(1,1,7)$	204	208	215
$(1,2,6)$	208	240	260
$(3,3,3)$	215	260	288

where the triples specify the class sizes of an operation. The finite enumeration covers all of them, including the ordering-dependent internal merges.

4.3 Thirteen cyclic plans

Take the ordered seed

$$(11,2,8,9,7) \mid (10,1,0,5) \mid (6,4,3,12)$$

and add d modulo 13 to every label for $0 \le d < 13$. The within-block orders remain the shifted orders; sorting them would define different plans. There are 33 masks per plan. The maximum-dot matrix is circulant, with first row

$$(1064, 647, 651, 713, 647, 709, 651, 651, 709, 647, 713, 651, 647).$$

Its row sum is 9,100. Hence $S = 118,300$ and

$$b_{13}^2 = \frac{56}{169}. \tag{10}$$

The checker exhausts 184,041 mask-pair dot products. The seed was found by an exploratory search. Its validity depends only on this explicit seed and the complete integer checks; no optimality or exhaustive seed search is claimed.

For the final six-group tail use $(0,1) \mid (2,3) \mid (4,5)$. Its final triangle has maximum squared mask norm 216 and each internal merge at most 204, giving $b_6^2 = 2 \cdot 216/(25 \cdot 6^2) = 12/25$.

4.4 Why the gain is not merely schedule slack

The earlier best three-plan nine-group certificate has $b^2_{9,\mathrm{old}} = 6608/18225$ [11]. In the same packing model, the root taper fraction satisfies $\eta_0 < 1/(20\log_2 9)$. Even pushing that certificate to its boundary leaves root coefficient at least its infimum

$$\kappa b_{9,\mathrm{old}}\sqrt{20\log_2 9} = 30.731244617454\ldots,$$

which is larger than our complete time coefficient. Thus our improvement cannot be reproduced solely by reclaiming the rational slack of that particular old root certificate. This is a statement about the displayed certificate model; it is not an algorithmic lower bound or a global optimization over all methods in the corpus.

5 A strict four-stage schedule: coefficient below 30.414

The new finite menus permit a better fully charged schedule than simply replacing the menus of the prior $9 \to 13 \to 7$ certificate. Use the nine-group affine menu, the thirteen-group cyclic menu, the same nine-group affine menu once more, and then a fixed six-group tail. The tail plan is $(0,1) \mid (2,3) \mid (4,5)$. Its final triangle has mask squared norm $4\cdot 4 + 8\cdot 25 = 216$, and its other merges have maximum squared norm 204. Thus its local coefficient satisfies $b^2_6 = 12/25$.

Let $\lambda = \lceil \log_2 s \rceil$ and define

d	r_d	η_d
0	9	$15773/10^6$
1	13	$9/800$
2	9	$4477/10^6$
$d \ge 3$	6	$(923/250000)(12/25)^{d-3}$.

Set $u_d = \lfloor \eta_d \lambda \rfloor$, $B = \lceil 6\log_2 \lambda \rceil$, $a_{-1} = 0$, $a_d = \sum_{j\le d}(u_j + B)$, and stop at $M \le H = \lceil s/\lambda^{3/4} \rceil$. Use relative padding slack λ^{-3} and retain the imported frozen creation-precision entry semantics. Finite small exceptions are handled by direct contraction, after checking all applicability gates.

The prefix sums are

$$\eta_0 + \eta_1 = 27023/10^6, \qquad \eta_0 + \eta_1 + \eta_2 = 63/2000.$$

The tail sums to $71/10000$, so total taper capacity is $193/5000$. The following exact integer inequalities certify all required logarithms:

$$9^{20000} < 2^{63399}, \quad 13^{2000} < 2^{7401}, \quad 9^{100} < 2^{317}, \quad 6^{200} < 2^{517}.$$

For the symmetric root gate $Q_0^2 s^{9/10}$, then the retained-side gates $Q_d s^{9/10}$, they give the respective fixed gaps below exponent one

$$\frac{7573}{10^{10}}, \qquad \frac{2777}{2\cdot 10^9}, \qquad \frac{29}{200000}, \qquad \frac{219}{10^6}.$$

All gaps are strict. In particular the thirteen-group logarithm is bounded by $7401/2000$, rather than the weaker $3701/1000$ which would not suffice for this schedule. The fresh-side degree term and packet-rounding term are bounded by leading exponent

$$\frac{193}{5000}\left(2\frac{7401}{2000} + 1\right) = \frac{1621393}{5000000} < 1.$$

New degrees have exponent at most $193/5000 + o(1) < 9/10$.

To close the depth argument without assuming padding in advance, consider hypothetical active tail depths at most $\frac{3}{4}\log_2 \lambda + O(1)$. The least retained taper is

$$\lambda^{\varepsilon+o(1)}, \qquad \varepsilon = 1 - \frac{3}{4}\log_2(25/12) > 0,$$

where $(25/12)^3 < 16$ is an exact certificate. The cumulative buffers are $O((\log\lambda)^2)$, and $s/M < \lambda^{3/4}$. Hence the fixed positive packing gaps absorb all guards and polylogarithmic losses uniformly in this hypothetical depth range. Padding is nearly lossless and the true tail child ratio is $1/3 + o(1) < 1/2$, closing the depth bootstrap. The product of relative padding errors over $O(\log\lambda)$ levels is $1 + o(1)$, and the floors are uniformly relative $o(1)$.

Every retained valuation satisfies $D \geq 2^{a_{d-1}+B+1}$. The inherited creation precision $O(s\lambda^2 2^{a_{d-1}})$ divided by MD/u_d is at most $O(\lambda^{15/4}2^{-B}) = o(1)$. Thus the same sharp local precision, weighted entropy, and entry-evaluation interfaces apply uniformly. The wire counts satisfy

$$M_0 \leq s, \quad M_1 \leq (1+o(1))\frac{2s}{9}, \quad M_2 \leq (1+o(1))\frac{4s}{117}, \quad M_d \leq (1+o(1))\frac{8s}{1053}\left(\frac{1}{3}\right)^{d-3} \quad (d \geq 3).$$

Write $K_d := \mathcal{K}(M_d, u_d)$ and $\alpha_r = \sqrt{2(r-1)}/r$. The local paid recurrence is

$$T_d \leq \mathrm{poly}(s)\left[2^{(\alpha_{r_d}+o(1))K_d}T_{d+1} + 2^{(b_{r_d}+o(1))K_d}\right],$$

with $b_9 = 1/\sqrt{3}$, $b_{13} = \sqrt{56}/13$, $b_6 = \sqrt{12}/5$. The singleton cache must be evaluated by the actual child contraction; its cost is the first term, not a unit-cost oracle. On the scale $s/\sqrt{\lambda}$, define

$$K_0^* = \kappa\sqrt{10^6/15773}, \qquad K_1^* = \kappa\frac{2}{9}\sqrt{800/9},$$
$$K_2^* = \kappa\frac{4}{117}\sqrt{10^6/4477}, \qquad K_3^* = \kappa\frac{8}{1053}\sqrt{250000/923}.$$

The tail ratio is $\gamma = (1/3)/\sqrt{12/25} = 5/(6\sqrt{3})$. Since $\alpha_6/(1-\gamma) > b_6$, unrolling the recurrence bounds every branch by the maximum of

$$\begin{aligned}
C_0 &= b_9K_0^* = 29.466030256594790\ldots,\\
C_1 &= \alpha_9K_0^* + b_{13}K_1^* = 30.413348726348439\ldots,\\
C_2 &= \alpha_9K_0^* + \alpha_{13}K_1^* + b_9K_2^* = 29.634547541824756\ldots,\\
C_\infty &= \alpha_9K_0^* + \alpha_{13}K_1^* + \alpha_9K_2^* + \frac{\alpha_6}{1-\gamma}K_3^* = 30.013332173441237\ldots.
\end{aligned}$$

All displayed comparisons are enclosed by rational intervals in the checker, rather than inferred from rounded floating-point values.

The resulting exact time and space coefficients are

$$\boxed{C_T = \frac{\pi}{\ln 2}\left(\frac{4}{9}\sqrt{\frac{2000000}{15773}} + \frac{160\sqrt{14}}{351}\right) = 30.413348726348439\ldots}$$

and

$$\boxed{C_S = \frac{\pi}{\ln 2}\sqrt{\frac{2000000}{47319}} = 29.466030256594790\ldots.}$$

The local storage coefficients after the root are respectively $7.73038806\ldots$, $1.89085996\ldots$, and $0.55525285\ldots$; they are smaller than C_S. Ancestor tables add in size, so their exponents are maximized, not added. Direct leaves cost $2^{O(s/\lambda^{3/4})}$, original entry evaluation costs $2^{O(s^{9/10})}$ times a polynomial, and new entry routines cost $\exp(s^{193/5000+o(1)})$; all are lower order. The product of fixed polynomial overheads over $O(\log\lambda)$ levels adds $O(\lambda\log\lambda) = o(s/\sqrt{\lambda})$ to the time exponent.

Thus, under the same explicit imported contraction interface, the improved algorithm uses deterministic time

$$(s+\mathcal{L})^{O(1)}2^{(C_T+o(1))s/\sqrt{\lambda}}$$

and space with the same form and coefficient C_S. The safe eventual coefficients 30.414 and 29.467 are certified. This strictly improves the inspected generic cached certificates $30.891018920366\ldots$ time and $30.753912832055\ldots$ space. It remains a coefficient improvement on the same logarithmic scale, not a proof of a stronger unrestricted SAT exponent order or an observed end-to-end runtime improvement.

6 Fast, precision-safe Cauchy feature application

6.1 A different local interface

The global theorem above reduces the cost of summing mode indices. This section improves a different step: evaluating all source linear forms at a given port assignment. Earlier packets give quadratic feature-table construction or quadratic streamed application [8, 12, 10]. The output of a full $N \times N$ table already has quadratic size. A matrix-vector product has only linear input and output size and need not materialize that table. We exploit this distinction.

Fast polynomial products, multipoint evaluation, transposed algorithms, and Toeplitz convolution are classical [15]. The contribution here is their exact realization for the corpus's normalized two-adic features, including the extra precision required by the normalization. We found no such implementation or argument in the inspected packet text. This is a corpus-relative finding, not a claim of priority for fast structured linear algebra in general.

Let $L = 2^a$, $N = L/2$, $R = 2^i$ with $0 \le i < a$, and define the increasing orientation slots

$$s_j = (j \bmod R) + 2R\,\lfloor j/R \rfloor \qquad (0 \le j < N).$$

Let T be a specified element of valuation i in the unramified coefficient ring, available at the increased precision required below. The source has $T = t + 2^{a+1}\zeta$ with a fixed canonical description, so such evaluation is available. For $\epsilon \in \{1,-1\}$ put $x_j = \epsilon s_j$, $z_b = \epsilon s_b$ and

$$E_b = 2\,\lfloor b/R \rfloor - \operatorname{popcount}(\lfloor b/R \rfloor).$$

The normalized features of [8] are

$$F_{jb} = 2^{-E_b}\frac{T}{T + x_j - z_b}\prod_{h<b}\frac{x_j - z_h}{T + x_j - z_h}. \tag{11}$$

They are integral and vanish for $j < b$. Positive sign gives the target features; negative sign gives the source features, commonly denoted G. Thus the source forms are precisely $G^{\mathsf{T}}v$ for the vector of real column bits. Omitted ports may be omitted from the sum or zero-filled. We allow the first $1 \le B \le N$ modes and any specified set J of real ports. Vector entries may belong to the finite free nilpotent extension as well. Canonical lifts of vector entries can be chosen freely: integrality of F makes the output modulo 2^p independent of those choices. This does not assert that canonical lifting is a ring homomorphism.

6.2 The denominator valuations

Let

$$D_B(X) = \prod_{h<B} (T + X - z_h), \qquad H = E_{B-1}.$$

The sequence E_b is nondecreasing. For $j \in J$, the known valuation of the denominator is

$$\kappa_j = v_2(D_B(x_j)) = \sum_{d=1}^{i} \#\{h < B : h \equiv j \pmod{2^d}\}. \tag{12}$$

Indeed the slot difference has valuation either below i or strictly above i; both slots have bit i equal to zero. Since $v_2(T) = i$, the summand denominator has valuation $\min(i, v_2(j-h))$, with i for $j = h$. Counting divisibility levels proves (12). For $\kappa = \max_{j\in J} \kappa_j$, with zero for an empty set,

$$\kappa \le B(1 - 1/R) + i, \qquad H \le 2(B-1)/R.$$

In particular,

$$H + \kappa \le 2B + i, \qquad H + 2\kappa \le 2B + 2i. \tag{13}$$

The implementation uses the exact valuations, not merely these upper bounds.

6.3 Synthesis by a rational Newton product tree

For a consecutive interval $I = [l, r)$ define

$$A_I(X) = \prod_{h\in I} (X - z_h), \quad B_I(X) = \prod_{h\in I} (T + X - z_h),$$

and, with $w_b = c_b 2^{H-E_b} T$,

$$C_I(X) = \sum_{b\in I} w_b \prod_{l\le h<b} (X - z_h) \prod_{b<h<r} (T + X - z_h).$$

For consecutive halves $I = I_1 \cup I_2$ the identity

$$C_I = C_{I_1} B_{I_2} + A_{I_1} C_{I_2} \tag{14}$$

is division free. At the root,

$$(Fc)_j = \frac{C_{[0,B)}(x_j)}{2^H D_B(x_j)}.$$

Build the product tree and evaluate the numerator and denominator at all x_j modulo $2^{p+H+\kappa}$. Monic product/remainder trees need only division by monic polynomials, so no nonunit leading coefficient is inverted. By integrality of F, the numerator at x_j is divisible by $2^{H+\kappa_j}$. Divide that power of two exactly, invert the unit part of the denominator, and reduce modulo 2^p. The guard leaves at least p correct bits. All unit inverses can be obtained by one batch inverse and linear multiplication work.

6.4 The transpose requires a separate guard

It would be invalid simply to transpose an algorithm that contains divisions by powers of two. We instead derive a division-free linear map and justify its final divisions directly.

Put $P = p + H + \kappa$. First build and evaluate D_B at precision $P_{\text{high}} = p + H + 2\kappa$. Its normalized units $u_j = D_B(x_j)/2^{\kappa_j}$ are then known modulo 2^P. Batch-invert them and set

$$w_j = 2^{\kappa-\kappa_j} u_j^{-1} v_j \pmod{2^P}.$$

For $N_b(X) = \prod_{0\le h<b}(X - z_h)\prod_{b<h<B}(T + X - z_h)$, compute $q_b = \sum_j N_b(x_j)w_j$ as follows. First obtain the first B moments

$$m_k = \sum_j w_j x_j^k$$

from the first B coefficients of $\sum_j w_j/(1 - x_j Z)$. A balanced numerator/denominator construction and power-series inversion do this. The series denominator has constant coefficient one. Newton inversion $g \leftarrow g(2 - fg)$ works in this ring; it does not require that two be invertible.

Next transpose the division-free recurrence (14). If d is the parent coefficient functional, its child functionals are

$$d_1[k] = \sum_t B_{I_2}[t]d[k + t], \qquad d_2[k] = \sum_t A_{I_1}[t]d[k + t].$$

Each is a polynomial correlation. At the leaves the resulting values are the q_b. Finally,

$$(F^{\mathsf{T}} v)_b = \frac{T q_b}{2^{\kappa+E_b}} \pmod{2^p}. \tag{15}$$

The numerator is divisible by the displayed power because F is integral. Since $E_b \le H$, the core precision P leaves at least p bits. The higher precision for denominator preparation is essential: normalizing a denominator first consumes κ_j bits before its unit inverse is used at precision P.

Theorem 6.1 (Fast guarded feature application)**.** *Let $m = B + |J|$. Both synthesis and the transposed source-form map can be computed without a dense feature table in*

$$O\big(\mathsf{M}_{\mathcal{A}}(m, p + O(B + i))\log(m + 2)\big)$$

arithmetic cost, plus linear unit arithmetic and a batch inverse, where $\mathsf{M}_{\mathcal{A}}$ is polynomial-multiplication cost over the actual coefficient algebra. A stored product tree uses $O(m\log(m+2))$ coefficient storage. The required precisions are respectively $p + H + \kappa$ and $p + H + 2\kappa$ for synthesis and transpose preparation.

Proof. The preceding identities prove correctness. At each balanced-tree level the total polynomial degree is $O(m)$; standard superlinear multiplication-cost bounds sum to the displayed complexity. Monic remainders, constant-one series inversion, and correlations have the same multiplication-based bounds. The storage bound counts the coefficients across tree levels. Equations (13) give the precision assertion. □

6.5 Odd interpolation nodes need only one convolution

When $i = 0$ the formula simplifies substantially: $s_j = 2j$ and T is a unit. Let

$$f_j = v_2(j!) = j - \operatorname{popcount}(j), \quad o_j = j!/2^{f_j}, \quad D_j = \prod_{h=0}^{j}(T + 2\epsilon h), \quad D_{-1} = 1,$$

and put $u_j = o_j/D_j$, $v_j = D_{j-1}/o_j$. All these denominators are units. Direct cancellation in (11) gives

$$F_{jb} = \epsilon^b T\, 2^{f_j - f_b - f_{j-b}} u_j v_{j-b} \qquad (j \geq b). \tag{16}$$

The exponent equals $v_2\binom{j}{b} \geq 0$. To verify the identity, the numerator product in (11) is $(2\epsilon)^b j!/(j-b)!$, the denominator product is D_j/D_{j-b-1}, and $E_b = b + f_b$.

Theorem 6.2 (One-convolution source forms)**.** *For all N modes and ports at $i = 0$, either Fc or $F^{\mathsf{T}}v$ is computable with one polynomial convolution, $O(N)$ preparation, and one batch inverse at precision $p + 2f_{N-1}$.*

Proof. Set $K = f_{N-1}$ and $V_t = 2^{K-f_t} v_t$ at that precision. For synthesis convolve V with $c'_b = c_b \epsilon^b T 2^{K-f_b}$. Multiply coefficient j by u_j and divide by 2^{2K-f_j}; equation (16) gives $(Fc)_j$. For the transpose correlate V with $a_j = v_j^{\text{input}} 2^{f_j} u_j$. At index b, multiply by $\epsilon^b T$ and divide by 2^{K+f_b}. Again (16) gives exactly the desired answer. One reversed polynomial product yields all correlations. Each division is exact and consumes at most $2K$ bits. The odd parts o_j and products D_j are prepared by recurrence and batch inversion in linear work. □

In the scalar ring, quasi-linear integer multiplication with Kronecker substitution gives $\widetilde{O}(N(p + N))$ bit operations, including the guard. It is therefore $\widetilde{O}(Np)$ when $p \geq N$. The coefficient algebra's rank must be included for extension rings, or one can leave the cost as $\mathsf{M}_{\mathcal{A}}$. The executable supplied here implements the scalar ring with CPython integers. It does not implement an optimal FFT backend or an optimized extension-ring version.

Taking $p = p_t$ at the active-node scale $p \asymp MD/u$ and $N = RD/2$ gives $p/N \asymp 2M/(Ru)$. Under the fixed schedule of this report, $R \leq 2^u = s^{\eta_d + o(1)}$, while active $M > s/\lambda^{3/4}$. Thus the extra $O(N)$ bits are lower order than the active precision. This argument must be rechecked for a different schedule; it is not a license to omit guards from a general complexity claim.

Finally, these transforms evaluate linear source forms. The subsequent nonlinear mode monomials and the paid child contractions still have to be summed. The transforms alone therefore leave the global exponent scale unchanged. Their benefit is a provable local asymptotic improvement and, for the odd-node specialization, a measured improvement on the supplied small kernel fixtures.

7 What the new source changes about the order question

The new cached coefficients improve a certified exponential constant. They leave the worst-case scale $n \log\log n/\sqrt{\log n}$ unchanged. The user-supplied Anthropic development introduces a different algebraic mechanism, and therefore deserves more than a comparison of its headline bound. This section examines a possible composition, makes one of the source's suggested extensions precise, and proves a restricted obstruction to a natural implementation of that extension.

7.1 A direct composition spends the apparent extra saving

The earlier precision packet gives an exact rank encoding of the structured mode sum [5]. If the first deformation uses r groups, its existing collision-packing certificate allows a retained level range of order

$$u \asymp \frac{\log M}{\log r}.$$

Its compressed mode-index graph consequently has a certified bit budget on the scale

$$B \asymp \frac{M}{\sqrt{u}} \asymp \frac{M\sqrt{\log r}}{\sqrt{\log M}}. \tag{17}$$

These are the scales of that proof's envelopes, not lower bounds on the actual entropy of every instance.

At fixed or polylogarithmic r, a group tensor can have degree comparable to B/r, so the original bounded-degree contraction theorem cannot simply be invoked again at size B. This missing interface is already identified in the earlier reports [5, 6]. Anthropic's complete-graph algorithm does permit arbitrary integer-valued node tensors with charged entry access [4, Theorem A]. It is therefore a more appropriate possible backend, after balancing and representation costs have been supplied.

However, the inspected backend has an exponent rate of order $e(r) \asymp 1/\sqrt{\log r}$ along its certified schedule. Even the optimistic calculation that leaves out balancing and entry-evaluation overheads gives

$$e(r)B \asymp \frac{1}{\sqrt{\log r}} \frac{M\sqrt{\log r}}{\sqrt{\log M}} = \frac{M}{\sqrt{\log M}}. \tag{18}$$

For fixed r the backend changes a constant. Increasing r spends the corresponding saving in the first-stage retained-level budget. Allowing r to grow would also require uniform accounting for the exact-weight arrays and the newly expensive group entries. Equation (18) is thus a diagnosis of this proposed composition, not a completed composition theorem or an algorithmic lower bound. The new source resolves an important admissibility mismatch but does not automatically supply the missing $M/\log M$ summation method.

7.2 A cost-complete form of the source's growth suggestion

Appendix B of the Anthropic paper builds crossing decompositions from operators on the graded space $V = \mathbb{Q}[t_1, t_2, \ldots]$, with $\mathrm{weight}(t_j) = j$. Here $\dim V_E = p(E)$ is the ordinary partition number. Its two operator orders pass through spaces of weights at most $2h + s$ and $2h - s$. The asymptotic relation $\log_2 p(E) = A\sqrt{E} + O(\log(E + 2))$ gives the rank-sum parameter $\rho = 1/\sqrt{2}$ and the logarithmic exponent $\log_2(1/\rho) = 1/2$ [4, Appendices A and B].

The source itself observes in Appendix H that growth like E^β would instead give $\rho = 2^{\beta-1}$. The following proposition formalizes that suggestion with the resource hypotheses needed for an algorithm. The growth-to-rate idea belongs to the source. No algebra meeting the stronger hypotheses below is constructed in this report.

Proposition 7.1 (Conditional graded-growth transfer)**.** *Fix a rational $0 < \beta < 1$. Suppose an effectively presented graded rational operator system $V = \bigoplus_{E \geq 0} V_E$ has fixed bases, finite, nondecreasing dimensions $d_E \geq 1$, and*

$$\log_2 d_E = AE^\beta + O(\log(E + 2)), \qquad A > 0. \tag{19}$$

For each bit length b, let h be the least integer with $d_h \geq 2^b$ and label 2^b distinct basis states of V_h by $x \in \{0, 1\}^b$. Assume the following additional properties, uniformly in b.

1. *A nondegenerate dual pairing and an exact operator identity give operators $Q_s(x, v)$ on V_h, for $|s| \leq 2h$, and a positive integer D_h, such that*

$$\sum_{s=-2h}^{2h} Q_s(x, v) = D_h[x = v] \, \mathrm{Id}_{V_h} .$$

2. *Each $Q_s(x, v)$ has two exact factorizations, respectively as a sum of $h^{O(1)}$ products $B_{v,s,j}A_{x,s,j}$ and as a sum of $h^{O(1)}$ products $\widetilde{A}_{x,s,j}\widetilde{B}_{v,s,j}$. The intermediate spaces in the first factorization*

have weights at most $2h + s$, and those in the second at most $2h - s$. All factorization entries are integral, after any specified common denominator clearing. This exact two-order identity is a hypothesis; a dimension estimate alone does not imply it.

3. *There are uniform algorithms computing h, labels, index lists, D_h, piece values and factorization entries in $2^{O(b)}$ bit operations. The number of pieces is $b^{O(1)}$, and all integers have $b^{O(1)}$ bits. In particular the hypotheses include the basis and rank-decoding costs and effective exact locality, wherever locality is used to prove the preceding identities.*

Then the system supplies pieces in Anthropic's Definition 7 with $\rho = 2^{\beta-1}$. Relative to its Theorem A and dense reduction, this gives, for fixed width k,

$$\mathrm{poly}(n, m)2^{O_k(n \log\log n/(\log n)^{1-\beta})}$$

time at arbitrary density and $\mathrm{poly}(n, m)2^{O_{k,D}(n/(\log n)^{1-\beta})}$ *when $m \le Dn$. No polynomial-space conclusion is part of this implication.*

Proof. For a piece define $A_s(x, y, u, v) = \langle e_u^*, Q_s(x, v)e_y\rangle$, where e_u^* is the dual basis vector. The first identity gives $\sum_s A_s = D_h[x = v][y = u]$. The first factorization separates (x, y) from (u, v), and the second separates (x, u) from (y, v). By monotonicity of d_E, their respective numbers of rank terms are at most $h^{O(1)}d_{2h+s}$ and $h^{O(1)}d_{2h-s}$.

The matching upper and lower estimates in (19), together with minimality of h, give

$$h = \Theta(b^{1/\beta}), \qquad Ah^\beta \le b + O(\log(b + 2)).$$

Indeed $d_{h-1} < 2^b$ supplies the latter inequality up to $A(h^\beta - (h - 1)^\beta) = O(1)$. Rounding the logarithms of the rank term counts to bits and using concavity of x^β gives

$$\begin{aligned} c_s^{\mathrm{row}} + c_s^{\mathrm{col}} &\le A\big((2h + s)^\beta + (2h - s)^\beta\big) + O(\log h) \\ &\le 2A(2h)^\beta + O(\log h) \\ &\le 2^{\beta+1}b + O(\log(b + 2)). \end{aligned}$$

Thus Definition 7 holds with $4\rho = 2^{\beta+1}$; its other conditions are precisely the effective hypotheses above. Finitely many small values of b can be absorbed by enlarging its fixed constant. Theorem A has logarithmic exponent $\log_2(1/\rho) = 1 - \beta$. For fixed rational β, the rational rate approximations required by that theorem are computable by integer power comparisons, so its stated consequences apply. □

For example, $\beta = 1/3$ would replace the square-root denominator by $(\log n)^{2/3}$. This is a sufficient-condition calculation, not an achieved improvement. Even $\log d_E = \Theta(E^\beta)$ is insufficient to identify the displayed ρ: unrelated constants in the upper and lower estimates would remain in the rank bound.

7.3 Why deleting oscillator modes does not realize it

A natural candidate retains only square-weight generators, $\mathbb{Q}[t_1, t_4, t_9, \ldots]$. Its attraction is the smaller partition growth. Its immediate defect is that the current's positive commutator series has the form

$$C(z) = \sum_{j\ge1} j^2 z^{j^2}, \tag{20}$$

up to a monomial shift or nonzero coefficient normalization. It cannot satisfy the polynomial-clearing locality used in Appendix B. The obstruction extends beyond this one choice.

Lemma 7.2 (Rational series cannot have unbounded support gaps)**.** *Let $C(z) = \sum_{n\ge 0} c_n z^n$ over a field be rational and not a polynomial. After a finite prefix, its nonzero coefficient support has bounded gaps.*

Proof. Write $Q(z)C(z) = P(z)$ with $Q(0) = q_0 \neq 0$ and $d = \deg Q$. For $n > \deg P$ the coefficients satisfy

$$q_0 c_n + q_1 c_{n-1} + \cdots + q_d c_{n-d} = 0.$$

If d consecutive coefficients in this tail were zero, induction would force every later coefficient to vanish. This would make C a polynomial. The case $d = 0$ already makes C a polynomial and is excluded. Therefore sufficiently late gaps have length less than d. □

The series (20) has arbitrarily long zero runs between consecutive squares and infinitely many nonzero terms, so it is not rational. In particular, no nonzero polynomial Q makes QC a polynomial. In the independent-current construction, clearing the two-variable commutator by a fixed power of $(\omega_1 - \omega_2)$ would, on taking its positive expansion and setting $z = \omega_2/\omega_1$, imply just such rationality. More explicitly, up to a monomial shift the two-sided scalar distribution is $C(z) - C(z^{-1})$. Multiplying its negative part by $(1 - z)^N$ can produce positive exponents only below N. Thus annihilation of the distribution forces all sufficiently high coefficients of $(1 - z)^N C(z)$ to vanish. The square-weight replacement therefore lacks the exact identity on which the claimed rank bounds would depend.

Proposition 7.3 (A growth obstruction for independent oscillators)**.** *Let $S \subseteq \mathbb{Z}_{>0}$ be infinite and be the nonzero coefficient support, apart from finitely many indices, of a rational nonpolynomial current series. Give the polynomial space*

$$V = \mathbb{Q}[t_n : n \in S]$$

the grading $\mathrm{weight}(t_n) = n$*. There is $c > 0$ such that, for all sufficiently large E,*

$$\sum_{j=0}^{E} \dim V_j \ge 2^{c\sqrt{E}}. \tag{21}$$

Consequently this model cannot satisfy a uniform exact-degree upper bound $\dim V_E \le \exp(O(E^\beta + \log(E+2)))$ *with $\beta < 1/2$.*

Proof. By Lemma 7.2, a fixed integer d bounds the gaps in S after a finite prefix. For sufficiently large B, at least $B/(2d)$ members of S lie in $[1, B]$. Independently choosing exponent zero or one on these generators produces at least $2^{B/(2d)}$ distinct monomials, every one of weight at most $B|S \cap [1, B]| \le B^2$. Take $B = \lfloor \sqrt{E} \rfloor$. This proves the cumulative, rather than pointwise, lower bound (21). On the other hand, the proposed exact-degree upper bound would imply

$$\sum_{j=0}^{E} \dim V_j \le (E+1) \exp\big(O(E^\beta + \log(E+2))\big),$$

contradicting (21) when $\beta < 1/2$. □

The independence in this proposition is not removed by taking a quotient that preserves a nondegenerate Heisenberg vacuum representation. Suppose a nonzero vacuum v_0 is annihilated by positive operators a_n, the negative operators a_{-n} raise weight by n, operators of either sign commute among themselves, and

$$[a_n, a_{-m}] = \delta_{nm} \gamma_n \,\mathrm{Id}, \qquad \gamma_n \neq 0 \quad (n \in S)$$

over a characteristic-zero field. In a relation among creation monomials of the same weight, apply $\prod_n a_n^{\alpha_n}$ for any exponent tuple α appearing there. Every different same-weight tuple has some smaller coordinate and is killed. The matching tuple leaves $\prod_n \alpha_n! \gamma_n^{\alpha_n} v_0 \neq 0$. Hence its coefficient in the relation is zero, and all creation monomials are independent. Such a quotient retains the growth obstruction.

These statements concern the specified independent-current or Heisenberg-vacuum model. They do not rule out different noncentral operator algebras, other exact recovery identities, compressed tensor representations, or faster SAT algorithms. In particular, they are not general lower bounds for vertex algebras or for SAT. Their useful conclusion is narrower: merely deleting oscillator modes, or imposing relations while retaining the displayed Heisenberg premises, does not implement the source's smaller-growth suggestion.

7.4 Further research questions

1. Can a noncentral graded operator algebra supply the pairing, two-order identity and effective factorization of Proposition 7.1 with $\beta < 1/2$? A candidate must escape the preceding Heisenberg premises while preserving exact recovery and uniform bit costs.

2. Can a different exact recovery method use a nonrational current? Truncating its commutator is insufficient unless the residual is proved to vanish in the actual counting ring and the required precision is fully charged.

3. Can the mode-rank and clique algorithms be composed with complete balancing and entry-query accounting to improve a coefficient or the memory tradeoff? An order claim must additionally overcome (18).

4. Which relations among the *actual* group tensors permit implicit summation without materializing their formal boundary tables? Existing inventory barriers concern a representation; they do not establish independence or noncancellation of the values in those tables [5, 6].

The first two questions seek a new algebraic ingredient. The latter two seek a representation that uses more than pair entropy. None of these required interfaces has been supplied here, so the proved positive improvement remains the smaller cached contraction constant and the exact kernel refinement established elsewhere in this report.

8 Validation, reproducibility, and exact scope

8.1 What was checked

The finite correlation certificates and the rational schedule are replayable with the Python standard library. The main checker exhausts every support pair: 7,056 dot products for the four affine plans and 184,041 for the thirteen cyclic plans. The six-group tail is checked as well. Integer power comparisons certify the logarithm bounds; rational interval arithmetic certifies the displayed constants and strict dominance of the depth-one time branch. In particular the final intervals are

$$C_T \in [30.413348726348439, 30.413348726348440],$$
$$C_S \in [29.466030256594790, 29.466030256594791].$$

An independently written checker reconstructs the plans and their supports using bit sets, recomputes the integer certificates and rational packing gates, and directly unrolls 93 levels of the paid recurrence

as a numerical cross-check. It does not import the primary checker. The finite unrolling is supplemental: the proof handles the infinite geometric tail exactly.

The regrouping test uses 12 deterministic fixtures with at most 12 active binary directed indices. It compares literal enumeration of global assignments with separately implemented singleton caches, balanced binary merges and the final three-block sum over $(\mathbb{Z}/64\mathbb{Z})[V]/(V^2)$. All 72 plan contractions agree. The fixtures cover both orientations, spanning paths, disconnected components, isolated scalar factors and zero factors. The nine fixtures without a zero factor all have nonzero outputs. The checker also counts every singleton entry call, verifies the count $\sum_v 2^{\deg(v)}$, and checks that the exact fifth-power selector minimizes the independently reconstructed operation costs. This is a finite check of regrouping and resource accounting, not an implementation of fast matrix multiplication.

For the local transform, 246 cases compare with independently formed exact rational feature matrices, covering 10,160 entries; 27 larger cases compare with the quadratic valuation-aware recurrence; and 96 odd-node cases check the convolution specialization. Both signs, truncated mode lists, missing ports, precision one, and the adjoint identity are covered. The denominator guard bounds are checked in these cases. These finite tests support the implementation; the product-tree and precision arguments in Section 6 establish the general identities.

All deterministic checks pass in normal Python and in `python -O`, with identical parsed output. Validation uses explicit exceptions rather than depending on assertions that optimization could remove. The archived results include both modes and a summary of the comparison.

8.2 Matched local measurements

The following scalar transpose measurements include preparation, denominator inversions, increased working precision, and output recovery. Every measured output was checked against the quadratic implementation. Each number is the median of three runs at $p = 4N$ and odd interpolation valuation. The code uses CPython integers and Kronecker substitution; these are small local measurements, not an end-to-end SAT benchmark.

N	Quadratic (s)	One convolution (s)	Ratio	General tree (s)
32	0.000407	0.000174	2.34×	0.002486
64	0.001923	0.000853	2.25×	0.011521
128	0.013941	0.007689	1.81×	0.091038
256	0.168149	0.057822	2.91×	0.656261
512	1.706872	0.418585	4.08×	6.034264

The general product-tree prototype is slower at all these sizes. Its asymptotic coefficient-operation bound remains useful, but these data do not establish a practical crossover. The observed speedup belongs to the odd-node specialization. The reference environment was Python 3.12.14 on Linux 6.18.55, x86_64, glibc 2.39. Raw samples and environment metadata are included so the measurements are not mistaken for portable constants.

8.3 Proof status and limitations

The newly proved parts are the correlation-certificate inequality, the explicit affine and cyclic menus, their integration into a fully charged four-stage recurrence, the precision-safe rational transforms and odd-node convolution, and the restricted sparse-current obstruction. The result uses the established interfaces of the inspected manuscripts: exact Hermite extraction, normalized mode factorization, balanced packing, entry semantics, SAT reduction, and the rational matrix multiplication backend. Those dependencies are stated in Section 3; this report is not an independent verification of every upstream claim.

The new contraction algorithm permits subexponential space. It does not improve the strongest polynomial-space bound, prove a smaller logarithmic order for unrestricted SAT, provide a finite-size crossover threshold, or contain an end-to-end SAT implementation. Neither this new proof nor its upstream dependencies were formalized or replayed in Lean here. The source review of the supplied Anthropic revision does not change that statement. The conditional graded-algebra proposition requires a new construction meeting all of its hypotheses.

8.4 Reproducing the archive

From the extracted directory, run:

```
python3 code/check_manifest.py
python3 code/run_checks.py
python3 code/build_report.py
```

The first command checks the delivered file hashes. The second replays the finite certificates, independent audits, regrouping and kernel tests in both Python modes. It requires only Python 3.10 or later. PDF rebuilding requires pdfLaTeX and the packages named in `report.tex`; the supplied figure is already present. To regenerate it, install Matplotlib and run `code/make_figures.py`. Optional fresh timings use `python3 code/fast_cauchy_transform.py -benchmark`; they need not match the archived timings. The exact proof obligations, imported dependencies and unproved targets are also recorded in `PROOF_STATUS.json`.

References

[1] Vladimir Reshetnikov, *ProveIt: SubexponentialSAT*, project index and 29 refinement packets, commit `a7ae5ab55f9627f5af7bf77443b3c34ef8ef1be4`. Immutable project tree. The companion provenance manifest supplies exact file hashes and paths.

[2] ProveIt commit `a7ae5ab55f`, *Place nine round-4 sub-exponential 3-SAT packets in SubexponentialSAT*, October 26, 2026 Pacific. Commit and placement record.

[3] OpenAI, *The Exponential Time Hypothesis is False*, October 25, 2026, upstream commit `7ef5c0cc5ca4b0e072b7a050fca09ab24bc6fa3a`. Pinned source directory. The unmodified ProveIt copy under `lib/openai-math/preprints/` was read, especially `build/quantitative-contraction.tex` and `build/quantitative-reduction.tex`.

[4] Claude (Anthropic), *A $2^{o(n)}$ Time Algorithm for 3-SAT*, paper and Lean source in `anthropics/formal-math`, PR 31, revision `1a6eee293b9e00cb432eef6018a21f523623a9af`. Pinned paper and formalization. Particularly Theorem A, Appendices B and H, `MainResults.lean`, `Solution.lean`, and `comparator.json`. Source inspected; the Lean build and kernel replay were not run for this report.

[5] *Sharper precision for subexponential 3SAT*, October 26, 2026, ProveIt `Research/3sat_precision_refinement/`. Pinned report source. Used for sharp precision, the exact mode-rank encoding, and the explicitly unproved stronger summation hypothesis.

[6] *Asymptotic improvement analysis for the October 2026 3-SAT preprint*, ProveIt `Research/3sat_asymptotic_analysis/`. Pinned report source. Also *Rectangular cached contraction*, pinned source, for scoped precision and explicit-table barriers.

[7] *Asymmetric packet balancing*, October 26, 2026, ProveIt `Research/eth_asymmetric_packet_balancing/`. Pinned report source. Used for the asymmetric packing gates and nearly lossless child sizes.

[8] *Joint Cauchy modes*, October 26, 2026, ProveIt `Research/eth_joint_cauchy/`. Pinned report source. Used for full-orientation normalized rational features, their integrality, and slope-four mode weights.

[9] *Adaptive bounds for subexponential SAT*, October 26, 2026, ProveIt `Research/sat_adaptive_bounds/`. Pinned report source. Includes the common weighted entropy interface and the best displayed generic polynomial-space coefficient.

[10] *Subexponential SAT contraction tree*, October 26, 2026, ProveIt `Research/subsat_contraction_tree/`. Pinned cached-tree proof. Used for matrix table merging and the coordinate/CRT arithmetic bridge.

[11] *Adaptive contraction trees with exact support certificates*, October 26, 2026, ProveIt `Research/sat_adaptive_tree_certificate/`. Pinned report source. The leading comparable cached coefficient and the earlier strict $9 \to 13 \to 7$ schedule come from this packet.

[12] *Adaptive exact kernels*, October 27, 2026 archive date, ProveIt `Research/eth_adaptive_exact_kernels/`. Pinned report source. Also *SAT frontier improvements*, pinned source, and the prefix theorem in *Subexponential SAT contraction tree*, pinned source. These are the reviewed quadratic feature-construction/application interfaces.

[13] *Density-aware SAT improvement*, pinned source, and *ETH density refinement*, pinned source. Both appear in ProveIt at the review commit.

[14] J. Alman, R. Duan, V. Vassilevska Williams, Y. Xu, Z. Xu and R. Zhou, *More Asymmetry Yields Faster Matrix Multiplication*, arXiv:2404.16349v2, 2024. `https://arxiv.org/abs/2404.16349v2`. Only the strict rational-field inequality $\omega < 12/5$ is needed here; no current-record claim is made.

[15] J. von zur Gathen and J. Gerhard, *Modern Computer Algebra*, third edition, Cambridge University Press, 2013. Classical polynomial product trees, monic division, fast series inversion and multipoint evaluation underlie the transform construction.